2.6 Effect rotation of the axes on moment of inertia

We have learnt that by parallel axis theorem, we can transfer axes parallelly. Now let us see what happens when the axes are rotated. Consider an area with respect to a coordinate system x-y with origin O. Consider a small elemental area dA at a distance (x,y) with respect to this coordinate system. By definition of moment of inertia,

$I_{xx} = \int y^2\, dA$, $I_{yy} = \int x^2\, dA$, and $I_{xy} = \int xy\, dA$

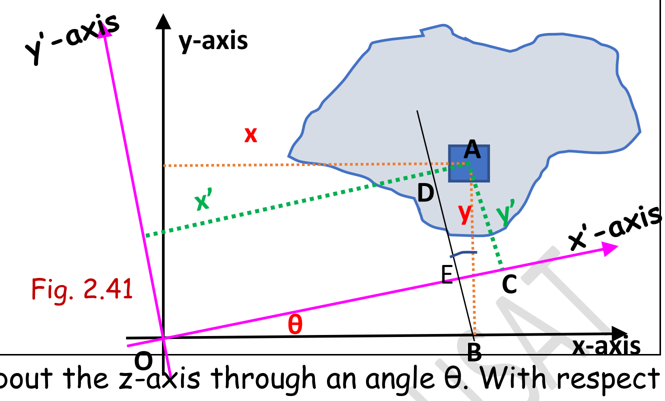


Fig. 2.41

Assume that the coordinate system is rotated about the z-axis through an angle θ. With respect to the new coordinate system x'-O-y', the elemental area is having coordinates (x', y'). Then $I_{x'x'} = \int y'^2\, dA$, $I_{y'y'} = \int x'^2\, dA$, and $I_{x'y'} = \int x'y'\, dA$. Draw a line BD⊥OC

We can write $x' = OC = OE + EC = OE + AD = x\cos\theta + y\sin\theta$ (from ΔOEB and ΔABD)

$y' = DE = DB - EB = y\cos\theta - x\sin\theta$ (from ΔABD and ΔOEB)

$$I_{x'x'} = \int y'^2\, dA = \int (y\cos\theta - x\sin\theta)^2\, dA = \cos^2\theta \int y^2\, dA + \sin^2\theta \int x^2\, dA - 2\sin\theta\cos\theta \int xy\, dA$$

$$I_{x'x'} = I_{xx}\cos^2\theta + I_{yy}\sin^2\theta - I_{xy}\sin 2\theta$$

$$I_{x'x'} = I_{xx}\left(\frac{1+\cos 2\theta}{2}\right) + I_{yy}\left(\frac{1-\cos 2\theta}{2}\right) - I_{xy}\sin 2\theta$$

$$I_{x'x'} = \left(\frac{I_{xx}+I_{yy}}{2}\right) + \left(\frac{I_{xx}-I_{yy}}{2}\right)\cos 2\theta - I_{xy}\sin 2\theta$$

$$I_{y'y'} = \int x'^2\, dA = \int (x\cos\theta + y\sin\theta)^2\, dA = \cos^2\theta \int x^2\, dA + \sin^2\theta \int y^2\, dA + 2\sin\theta\cos\theta \int xy\, dA$$

$$I_{y'y'} = I_{xx}\sin^2\theta + I_{yy}\cos^2\theta + I_{xy}\sin 2\theta$$

$$I_{y'y'} = I_{xx}\left(\frac{1-\cos 2\theta}{2}\right) + I_{yy}\left(\frac{1+\cos 2\theta}{2}\right) + I_{xy}\sin 2\theta$$

$$I_{y'y'} = \left(\frac{I_{xx}+I_{yy}}{2}\right) - \left(\frac{I_{xx}-I_{yy}}{2}\right)\cos 2\theta + I_{xy}\sin 2\theta$$

$$I_{x'y'} = \int x'y'\, dA = \int (x\cos\theta + y\sin\theta)(y\cos\theta - x\sin\theta)\, dA$$

$$= \sin\theta\cos\theta \int y^2\, dA - \sin\theta\cos\theta \int x^2\, dA + (\cos^2\theta - \sin^2\theta)\int xy\, dA$$

$$I_{x'y'} = I_{xx}\frac{\sin 2\theta}{2} - I_{yy}\frac{\sin 2\theta}{2} + I_{xy}\cos 2\theta = \left(\frac{I_{xx}-I_{yy}}{2}\right)\sin 2\theta + I_{xy}\cos 2\theta$$

We have defined the principal axes of inertia as the axes about which the product of inertia is zero. We can find out the orientation of the principal axes by setting $I_{x'y'} = 0$

$$\left(\frac{I_{xx}-I_{yy}}{2}\right)\sin 2\theta + I_{xy}\cos 2\theta = 0$$

$$\tan 2\theta = \frac{2I_{xy}}{I_{yy}-I_{xx}}$$

The moment of inertia with respect to the principal axes are known as the principal moments of inertia.

#Exercise 1

Prove that the moment of inertia will be either maximum or minimum with respect to the principal axes.

Q 1. Determine the orientation of the principal axes for the angle section shown and also find the principal moments of inertia.

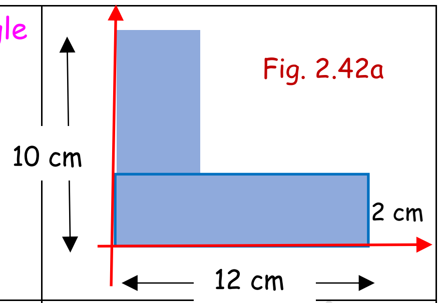


Fig. 2.42a

Dividing the angle section in to two rectangles

$$A_1 = 12 \times 2 = 24\ cm^2$$
$$A_2 = 8 \times 2 = 16\ cm^2$$

$$I_{1xx} = \frac{12 \times 2^3}{3} = 32\ cm^4 \qquad I_{2xx} = \frac{2 \times 8^3}{12} + 16 \times 6^2 = 661.3\ cm^4$$

$$I_{1yy} = \frac{12^3 \times 2}{3} = 1152\ cm^4 \qquad I_{2yy} = \frac{2^3 \times 8}{12} + 16 \times 1^2 = 21.3\ cm^4$$

$$I_{1xy} = 0 + 24 \times 1 \times 6 = 144 cm^4 \qquad I_{2xy} = 0 + 16 \times 1 \times 6 = 96\ cm^4$$

$$I_{xx} = I_{1xx} + I_{2xx} = 693.3 cm^4 \qquad I_{yy} = I_{1yy} + I_{2yy} = 1173.3\ cm^4$$

$$I_{xy} = I_{1xy} + I_{2xy} = 240\ cm^4$$

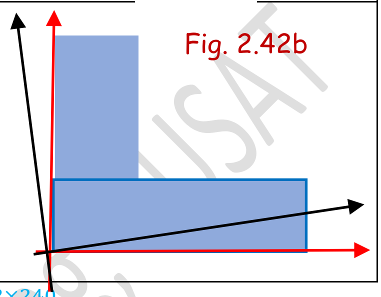


Fig. 2.42b

The orientation of the principal axes is given by $tan2\theta = \frac{2I_{xy}}{I_{yy}-I_{xx}} = \frac{2 \times 240}{1173.3-693.3} = 1$

$$2\theta = 45^0 \quad \ggg\gg \quad \theta = 22.5^0$$

The principal moments of inertia

$$I_{x'x'} = \left|\left(\frac{I_{xx}+I_{yy}}{2}\right) + \left(\frac{I_{xx}-I_{yy}}{2}\right) cos2\theta - I_{xy}\ sin2\theta\right|_{\theta=22.5} = \left(\frac{693.3+1173.3}{2}\right) + \left(\frac{693.3-1173.3}{2}\right) cos45 - 240\ sin45$$

$$= 593.88\ cm^4$$

$$I_{y'y'} = \left|\left(\frac{I_{xx}+I_{yy}}{2}\right) - \left(\frac{I_{xx}-I_{yy}}{2}\right) cos2\theta + I_{xy}\ sin2\theta\right|_{\theta=22.5} = \left(\frac{693.3+1173.3}{2}\right) - \left(\frac{693.3-1173.3}{2}\right) cos45 + 240\ sin45$$

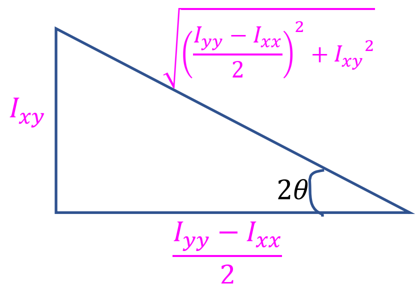
$$= 1272.71\ cm^4$$

The orientation of the principal axes is given by

$$tan2\theta = \frac{2I_{xy}}{I_{yy} - I_{xx}}$$

If we construct a right angled triangle as shown

$$cos2\theta = \frac{\frac{I_{yy}-I_{xx}}{2}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}} \quad \text{and } sin2\theta = \frac{I_{xy}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}}$$



$$I_{x'x'} = \left(\frac{I_{xx}+I_{yy}}{2}\right) + \left(\frac{I_{xx}-I_{yy}}{2}\right) cos2\theta - I_{xy}\ sin2\theta$$

$$= \left(\frac{I_{xx}+I_{yy}}{2}\right) + \left(\frac{I_{xx}-I_{yy}}{2}\right) \frac{\frac{I_{yy}-I_{xx}}{2}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}} - I_{xy} \frac{I_{xy}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}} = \left(\frac{I_{xx}+I_{yy}}{2}\right) - \frac{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2}{\sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}} - \frac{(I_{xy})^2}{\sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}}$$

$$I_{x'x'} = \left(\frac{I_{xx} + I_{yy}}{2}\right) - \sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}$$

$$I_{y'y'} = \left(\frac{I_{xx}+I_{yy}}{2}\right) - \left(\frac{I_{xx}-I_{yy}}{2}\right) cos2\theta + I_{xy}\ sin2\theta$$

$$= \left(\frac{I_{xx}+I_{yy}}{2}\right) - \left(\frac{I_{xx}-I_{yy}}{2}\right) \frac{\frac{I_{yy}-I_{xx}}{2}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}} + I_{xy} \frac{I_{xy}}{\sqrt{\left(\frac{I_{yy}-I_{xx}}{2}\right)^2 + I_{xy}^2}} = \left(\frac{I_{xx}+I_{yy}}{2}\right) + \frac{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2}{\sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}} + \frac{(I_{xy})^2}{\sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}}$$

$$I_{y'y'} = \left(\frac{I_{xx} + I_{yy}}{2}\right) + \sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + I_{xy}^2}$$

Hence, the principal moments of inertia can be written as

$$I_{11,\ 22} = \left(\frac{I_{xx} + I_{yy}}{2}\right) \pm \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + {I_{xy}}^2}$$

Q 2. Determine the orientation of the principal axes for the section shown and also find the principal moments of inertia with respect to its centroid.

Refer Q.2 in section 2.2 and Q.6 in section 2.5

From the left extreme and bottom edges, the centroid is at

$$\bar{x} = \frac{(12\times2)\times14 + (6\times2)\times9 + (10\times2)\times5}{20+12+24} = 9.71\ cm$$

$$\bar{y} = \frac{(12\times2)\times1 + (6\times2)\times5 + (10\times2)\times9}{20+12+24} = 4.71\ cm$$

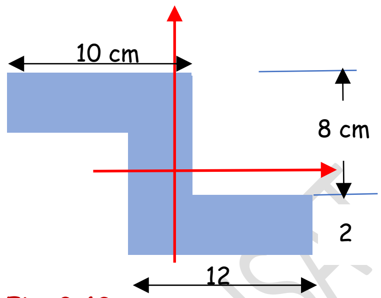


Fig. 2.43a

$$I_{1xx} = \frac{12\times2^3}{12} + (12\times2)(4.71-1)^2 = 338.33 \qquad I_{1yy} = \frac{12^3\times2}{12} + (12\times2)(14-9.71)^2 = 729.69$$

$$I_{2xx} = \frac{2\times6^3}{12} + (6\times2)(5-4.71)^2 = 37 \qquad I_{2yy} = \frac{2^3\times6}{12} + (6\times2)(9.71-9)^2 = 10.05$$

$$I_{3xx} = \frac{10\times2^3}{12} + (10\times2)(9-4.71)^2 = 374.75 \qquad I_{3yy} = \frac{10^3\times2}{12} + (10\times2)(9.71-5)^2 = 610.34$$

$$I_{1xy} = 0 + (12\times2)(14-9.71)(1-4.71) = -382$$

$$I_{2xy} = 0 + (6\times2)(9-9.71)(5-4.71) = -2.47$$

$$I_{3xy} = 0 + (10\times2)(5-9.71)(9-4.71) = -404.12$$

$$\boldsymbol{I_{xx} = 750\ cm^4} \qquad \boldsymbol{I_{yy} = 1350.08\ cm^4} \qquad \boldsymbol{I_{xy} = -788.59\ cm^4}$$

The orientation of the principal axes is given by $tan2\theta = \frac{2I_{xy}}{I_{yy}-I_{xx}} = \frac{2\times(-788.59)}{1350.08-750} = -2.62$

$$2\theta = -69^0 \quad \ggg \quad \theta = -34.5^0$$

The principal moments of inertia can be found out by

$$I_{11,\ 22} = \left(\frac{I_{xx}+I_{yy}}{2}\right) \pm \sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 + {I_{xy}}^2}$$

$$= \left(\frac{750+1350}{2}\right) \pm \sqrt{\left(\frac{750-1350}{2}\right)^2 + (-788.59)^2} = 1050 \pm 843.7$$

$I_{11} = 1893.7\boldsymbol{cm^4}$, $\qquad I_{22} = 206.3\boldsymbol{cm^4}$

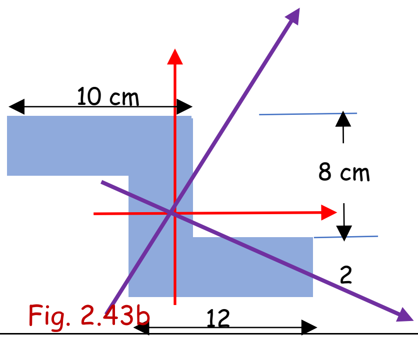


Fig. 2.43b

#Exercise 1

Calculate the orientation of the principal axes and the principal moments of inertia of the given shape with respect to its centroidal x-y axes.

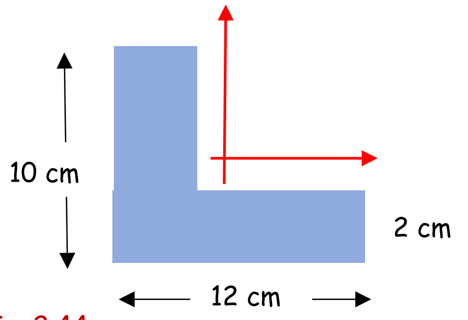


Fig. 2.44

#Exercise 2

Calculate the orientation of the principal axes and the principal moments of inertia of the given shape with respect to the given x-y axes.

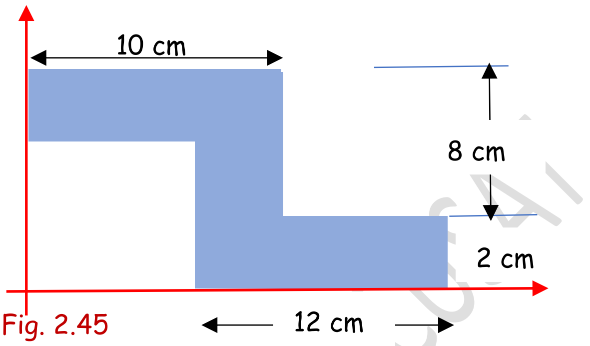


Fig. 2.45

#Exercise 5

Calculate the orientation of the principal axes and the principal moments of inertia of the I section with respect to the x-y axis.

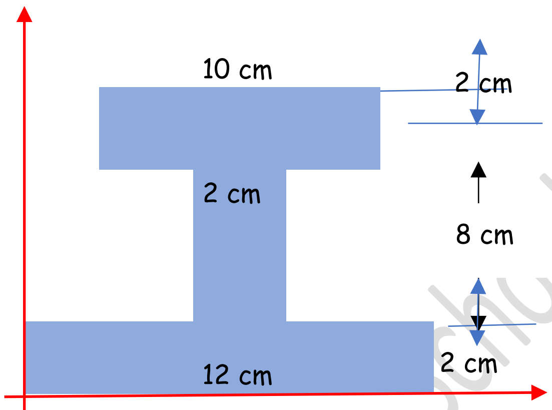


Fig. 2.46

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